ASUS UGEN AI LEAGUE 2026 · BATTLEFIELD LIGHTNING · WORKPLACE AI

VESTIGE

The world leaves a trace.

Flags worker–vehicle near misses on the cameras a construction site already has, and answers what happened, citing the footage. Built for the 2.5 W ASUS UGen300.

Stage I proposal · Wei-Lun Hsiao · B13902049 · National Taiwan University



AI-generated illustration

CLOSER TO HOME · NTU CAMPUS

Your bike was here.

Now it's gone.

08:52 · RECORDED



B-17 · LAST SEEN ?? · NOW UNKNOWN

AI-generated illustration

THE INVESTIGATION TODAY

The cameras saw it. *Finding it is another story.*

01 · ASK
Find out which cameras might cover the rack — and who owns them.

02 · REQUEST
Wait for footage exports from each owner, one by one.

03 · SCRUB
Watch hours of video to find a few seconds.

04 · STITCH
Piece together what happened across cameras — by hand.

≈29,000

bicycles on NTU's campus. In Sept 2024 NTU warned of "multiple recent cases of unlocked bicycles being ridden away."

59 min

for trained police reviewers to go through 18 minutes of CCTV — over 3× real time.

< 1 week

is how long private CCTV in Taiwan often keeps footage, sometimes only 2–3 days.

Sources: Liberty Times 2021-02-04; NTU notice (politics.ntu.edu.tw, 2024-09-13); West Midlands PCC 2020, citing Davis et al. 2018; MOJ Academy for the Judiciary

BIGGER STAKES, SAME QUESTION

This problem goes beyond missing objects.



TAIPEI METRO · 19 DEC 2025

The first alarm (17:24) came from smoke detectors, not cameras. When the footage matched the suspect (17:39), **he was already back at his hotel.** Afterwards, investigators reviewed more than 3,000 cameras.



WORK SITES · TAIWAN 2025–26

105 construction deaths in 2025 — 42% of all workplace fatalities. In July 2026 a reversing road roller ran over a worker standing behind it.

In both, the evidence existed — scattered across cameras, unlabeled, hard to search.

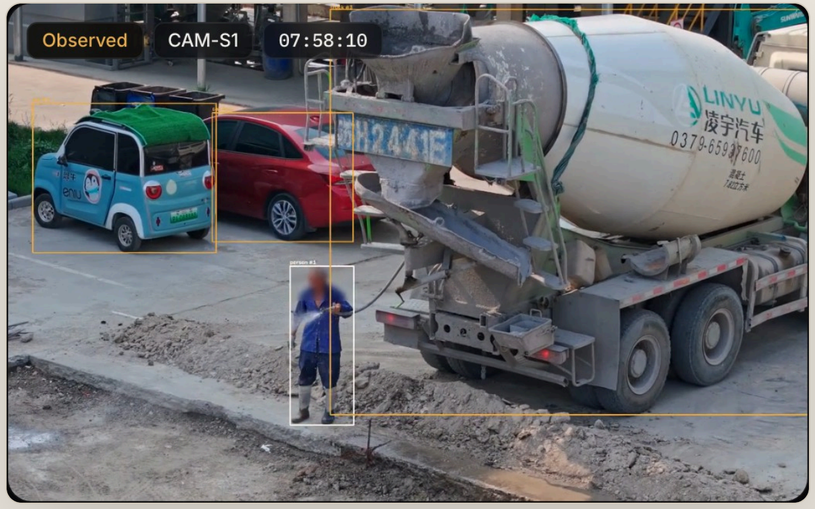
Sources: Taipei Metro; Taipei District Prosecutors; CNA; Ministry of Labor (A3) Illustrative imagery — not incident photos

PROBLEM DEFINITION → OUR TASK

Flag worker–vehicle near misses.
Answer what happened, with the footage.

The problem. Construction accounted for 42% of Taiwan's workplace deaths in 2025. Sites already run cameras, but no one watches them all day: near misses between workers and machines go unnoticed, and after an incident, finding the right seconds takes hours.

USERS	Safety officers and site managers · first pilot: WanZe
INPUT	Existing site cameras · files today, live streams in Stage II
FLAGS	A worker overlapping a vehicle in the image, and for how long
OUTPUT	Clip, duration, reason → a person confirms or dismisses → audit log
ASK	“Why was the 07:58 alert flagged?” → an answer that cites the clip
RUNS ON	Detection on the 2.5 W UGen300 · memory and Q&A on the host
MEASURED BY	At WanZe in Stage II: near misses caught, false alerts per hour, time to find evidence, FPS and power
TODAY	Laptop prototype. Overlap is measured in the image, not in metres; machinery is detected as a general “truck” class.



REAL PROTOTYPE OUTPUT · 07:58 ALERT

“Flagged because a worker overlapped the box of a truck for 15.0 s. Measured in the image, not as a real distance.”

Source: Ministry of Labor, 15 Jan 2026. Prototype on stock footage, faces blurred; UGen300 runs start in Stage II.

INTRODUCING

VESTIGE

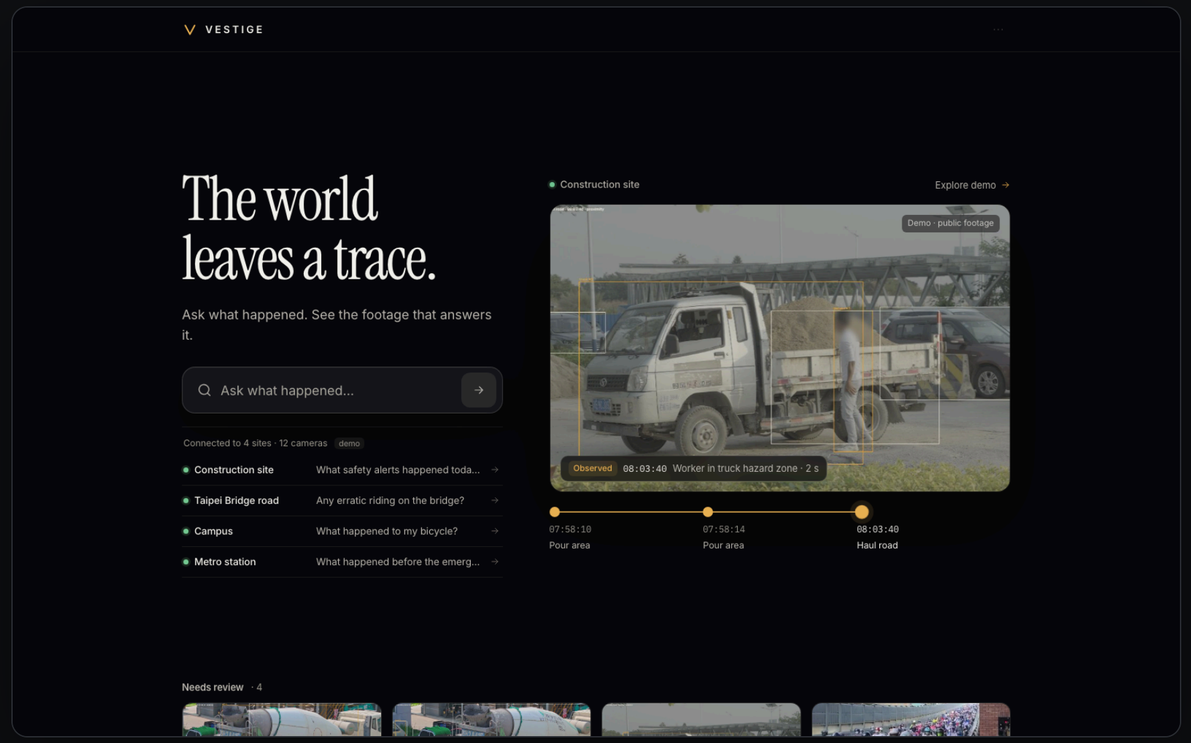
The world leaves a trace.

An edge AI memory for the physical world. Ask what happened in plain language — VESTIGE answers from what your cameras recorded and shows you the footage behind every sighting.

REMEMBER
objects and events, with provenance

INVESTIGATE
cited answers, gaps stay gaps

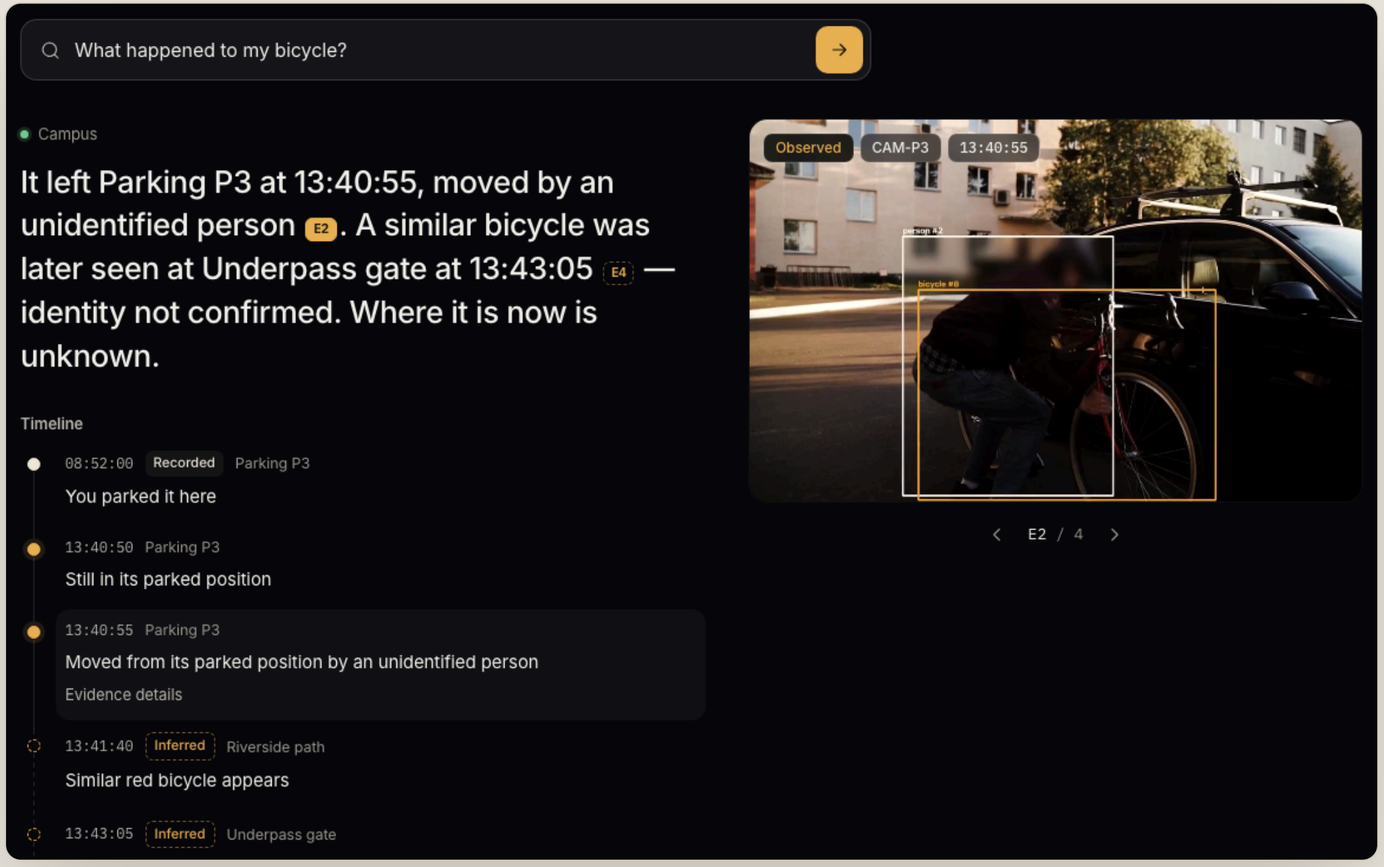
VERIFY
stress-tests its own detector



Working prototype · one search box across every connected site · demo data from public footage

Screenshot of the running prototype (laptop), 11 Oct 2026.

Ask once. Get the answer, *the footage and the gaps.*



1 · Conclusion first

"It left Parking P3 at 13:40:55, moved by an unidentified person … where it is now is unknown."

2 · Evidence that plays

Every sighting opens the processed clip at that second — faces blurred.

3 · Observed ≠ inferred

Cross-camera links use camera layout, travel time (39 s, 81 s) and colour (0.59): candidates, never an identity.

4 · Gaps stay gaps

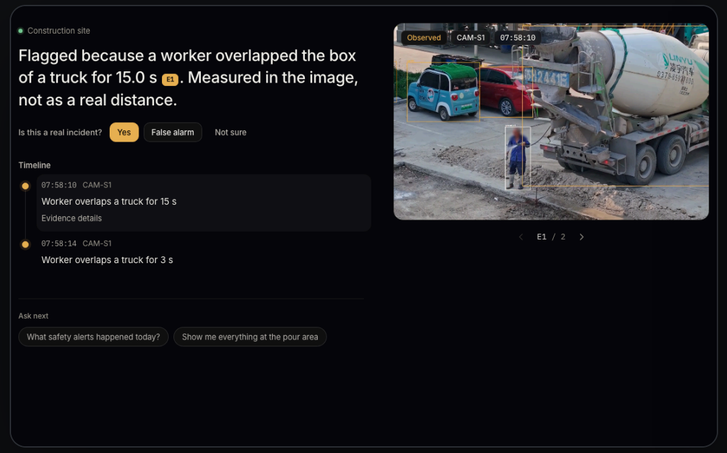
No camera beyond the underpass, so the answer ends in "unknown" instead of a guess.

Real prototype output · public stock footage re-timed into a demo campus · the owner check-in is staged.

ONE ENGINE · FIVE SITES

One search box. *Every site you connect.*

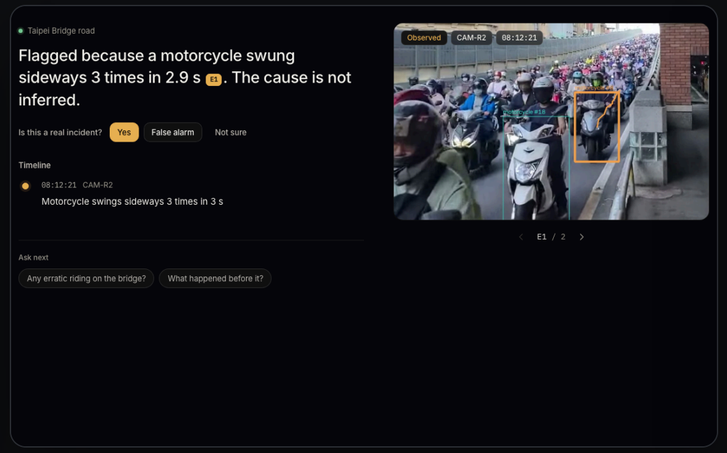
VESTIGE works out which site a question is about — from its words, the clock time of an alert, or wherever the cited evidence is — and labels the answer with that site.



CONSTRUCTION · FIRST MARKET

“Why was the 07:58 alert flagged?”

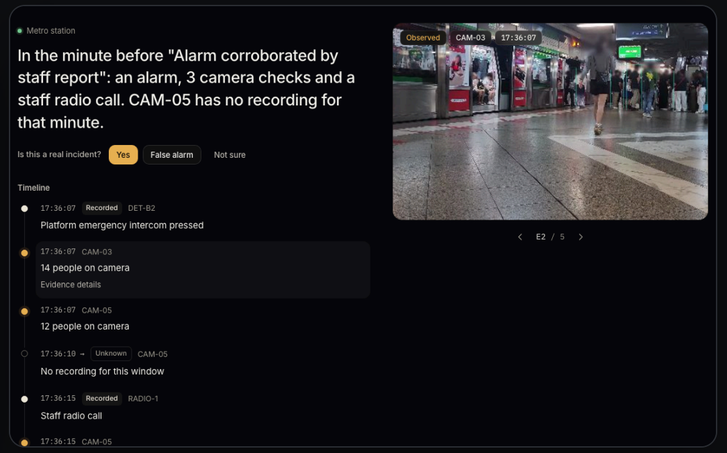
Every worker–truck overlap flagged with its clip and duration, explained (measured in the image), confirmed or dismissed by a person — and logged.



ROAD

“Any erratic riding on the bridge?”

A rider repeatedly moving sideways across the lane — the cause is not inferred.



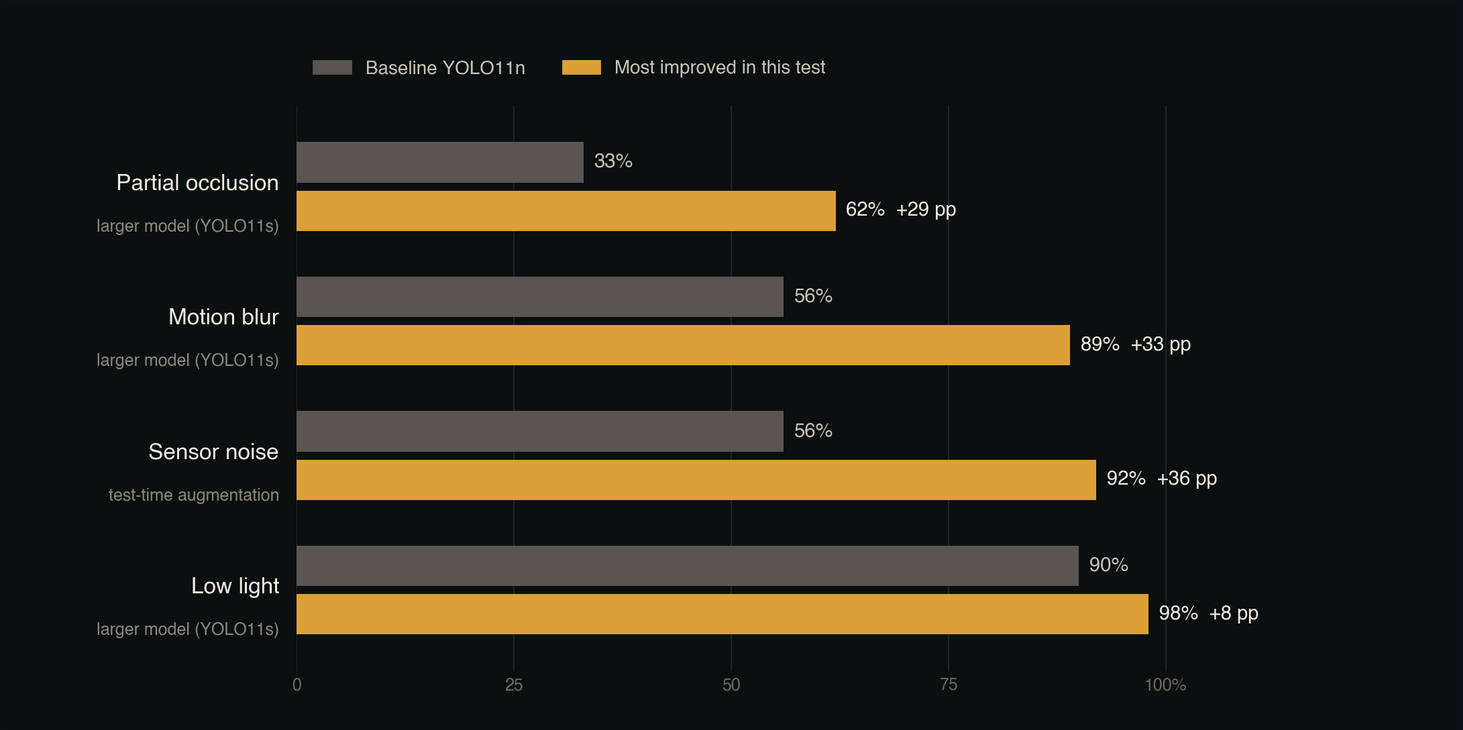
METRO STATION

“What happened before the alarm?”

Alarm, staff radio and cameras fused into one incident; a missing recording stays a gap.

Plus the campus bicycle story (p.07) and a records-only case (19 Dec 2025, published documents, no CCTV). Shared core: perception → World Memory → tool planner → cited answer. Demo sites use public footage and staged inputs.

Know where the AI fails — *and what helps.*



33% → 62%

Under partial occlusion the detector keeps a third of what it finds on clean frames. In this test a larger model raised that to 62% — at 75 vs 46 ms per frame.

And what hurts: naive low-light brightening lowered retention from 90% to 79%. We measure before we trust.

24 frames of Taipei Bridge traffic · 98 reference detections · laptop benchmark, 11 Oct 2026. Retention = share of the model's own clean-frame detections still found (IoU ≥ 0.5), not accuracy against human labels.

SYSTEM

How VESTIGE works.

1 · SOURCES

Authorized cameras
(video files today;
RTSP in Stage II)
Owner check-ins
Alarm I/O · radio

2 · PERCEIVE

YOLO + ByteTrack
Whisper on voice
VLM on demand
ASUS UGen300

3 · REMEMBER

Objects · tracks ·
zones
Events · relations
Evidence +
provenance

4 · INVESTIGATE

Agent with tools:
query · timeline ·
evidence · associate

5 · ANSWER

Search · timeline
Evidence viewer
Alerts to review · API ·
CLI

⟲ VERIFY · IncidentLab replays stored events under controlled corruptions → failure cases → better thresholds

Facts come from tools. The LLM plans queries and writes explanations; timestamps, places and frames come only from the memory store.

Cheap always, expensive rarely. Detection runs on every frame; the VLM wakes only for an event that needs a second look.

One modular monolith. Python/FastAPI, SQLite + media store, a web client; 2D zones now, 3D-ready schema.

DIFFERENTIATION

Why this is technically different.

Incumbents lead on scale and already search video in plain language. Our bet is a **combination**:

Capability (public material, Oct 2026)	Verkada	VIVOTEK VORTEX	Ambient.ai Pulsar	Voxel / Protex	VESTIGE today
Enterprise installed base	Yes	Yes	Yes	Yes	None yet
Natural-language search	Yes	Yes	Yes	Protex: yes	Prototype · built
Cross-camera timeline	Yes	Event insight	Preview	Workflow	Built
Explicit evidence states (observed / inferred / unknown)	Not found	Not found	Not found	Not found	Built
Whole pipeline on a 2.5 W accelerator	Camera + cloud	Cloud	GPU appliance	Edge GPU	Target, Stage II
User-run reliability stress tests	Not found	Not found	Not found	Not found	Built (IncidentLab)

Memory, not clips. Objects and events persist across cameras and days.

Answers you can audit. Every sighting cites a frame; gaps stay gaps.

Trust you can measure. IncidentLab shows where perception breaks.

"Not found" = absent from public product material, not proof of absence. Sources: Appendix C.

HARDWARE · BATTLEFIELD LIGHTNING

Designed for real-world deployment.



ASUS UGen300 · Hailo-10H
40 TOPS INT4 · 8 GB · 2.5 W

Role	Model	Hailo-10H, published	Laptop, measured	On UGen300
Detect, every frame	YOLO11n / v8s	302 / 166 FPS	47 ms / frame	Stage II
Listen, on voice	Whisper	25 tok/s (Base)	1.6 s / 8.7 s call	Stage II
Verify, per event	Qwen2-VL-2B	7 tok/s, 1 s TTFT	2.8 s / event	Stage II
Answer, on question	1.5B LLM	7.4 tok/s	2–4 s / answer	Stage II

Why Lightning, not Thunderstorm. The workload is event-driven: cheap detection always, 2B-class models rarely. It fits one 2.5 W stick; a 290 W GPU would mostly idle.

One chip, shared. All models share one HailoRT VDevice. Target: 8 cameras at 15 fps ≈ 40% of published YOLO11n capacity.

Not yet measured on a UGen300 — USB throughput, power and multi-model contention are the first Stage II tests.

Published: Hailo Model Zoo v5.4 (PCIe) and GenAI Zoo. Laptop: Apple M4 Pro, same models and input sizes.

STATUS, STATED PLAINLY

What runs today — and what Stage II adds.

Capability	Status	Notes
Detection + tracking + stitching on video	Built · tested	YOLO11 + ByteTrack on a laptop
World Memory with evidence states and provenance	Built · tested	SQLite; recorded / observed / inferred / unknown
Cited answers, one search box across sites	Built · tested	deterministic tool planner + site routing
Cross-camera candidate links	Built · tested	topology + travel time + colour, never identity
Construction overlap alerts + human review	Built	image-space, uncalibrated cameras
Road erratic riding · metro alarm/radio fusion	Built	TRACE replays; staged alarm and radio
IncidentLab stress tests + mitigations	Built · measured	24-frame laptop benchmark
Models on the UGen300 (Hailo-10H)	Stage II	HailoRT 5 port written, not yet run on hardware
Live RTSP cameras, user roles, retention	Stage II	file sources today

38 tests
automated: retrieval, citations, routing, isolation, upload, lab

1.6–3 ms
per investigation query on the laptop

409 → 9.4
alarms/hour → high-priority cards (synthetic stress test)

Built = runs in the repository today · tested = covered by automated tests · Stage II = needs the UGen300 or live sites.

GO-TO-MARKET

Our first market: construction.

WHY HERE FIRST

Cameras are already on site, hazards are physical and recurring, and every serious incident needs an investigation. Regulators are moving: Hong Kong now requires a Smart Site Safety System on public works above HK$30M.

WHO

Buyer: safety (EHS) and operations leadership. **Daily users:** site managers and safety officers who must answer "what happened, and why was it flagged?"

VALIDATION SIGNAL

Voxel, Protex AI and Intenseye sell vision safety; TSMC uses AI CCTV on its own construction site. The need is proven; edge-native, evidence-graded investigation is our opening.

Business model	Hypothesis to validate
Edge box	Host + UGen300 per site cluster
Subscription	Per camera, per year: memory, investigation, IncidentLab
Pilot → paid	Fixed-scope pilot, then site-wide roll-out
Channel	Camera vendors and system integrators

COST ANCHOR

Continuous cloud video analysis ≈ **US$353 per camera-month** (AWS list price). One 2.5 W UGen300 targets ~8 cameras.

FIRST CUSTOMERS

WanZe pilot, then Greater-Taipei contractors with camera-equipped active sites.

Sources: HK DEVB (4S); TechNews 2026-01-22; AWS pricing page (accessed 2026-10-10). Pricing is a hypothesis.

INDUSTRY VALIDATION

First pilot site agreed.

WanZe Construction (萬澤建設) has agreed to be our first pilot site for AI construction-site intelligence, with on-site testing planned for Stage II. Scope and terms are still to be defined together.

PILOT SITE · AGREED | SCOPE · TO DEFINE | DEPLOYMENT · NOT YET | RESULTS · NONE CLAIMED

Pilot plan	Proposed, to be confirmed jointly with WanZe
Users	Site managers and safety officers
Core use	Worker–machine proximity detection and incident investigation
Environment	Authorized site cameras and agreed test data only
Metrics	Event detection, time to find evidence, false alerts per hour, IncidentLab failure report
Next step	Agree scope, site access and the Stage II test plan

No paid contract, deployment or measured safety result is claimed. Company logo, site imagery and names used only with permission.

ROADMAP

From prototype to deployment.

NOW · OCT 2026

VESTIGE prototype on a laptop: four demo sites and a records-only case, cited answers, IncidentLab measured; first pilot site agreed with WanZe.

STAGE II · NOV 6 – DEC 4

Port detection, speech and VLM to one UGen300 and measure FPS, latency and power; live cameras; staged bike-lot scenarios; first on-site test at WanZe.

PILOT · 2027

WanZe site pilot, subject to agreed scope: proximity events, investigation, failure reports.

PLATFORM

Warehouses, logistics, labs, transit — same engine, new rule packs.

Stage II target	Today (laptop prototype)	How we will measure it
≥ 8 cameras at 15 fps per UGen300	~21 fps, one stream, M4 Pro	FPS, latency, power on the stick over USB
Event → evidence card < 2 s	Fusion < 1 ms; clips rendered offline	End-to-end timing on live streams
Evidence found ≥ 50% faster	Study designed, not run	10 staged events: manual vs VESTIGE
Failure modes documented	2 domains × 5 corruptions	IncidentLab report per scenario

Targets, not results. The user study is planned, not yet run.

FOUNDER



Wei-Lun Hsiao

Founder of Archean · Physical AI Researcher

Computer Science, National Taiwan University

RESEARCH	Offline, goal-conditioned reinforcement learning — Embodied AI Lab, NTU (2025–).
ALGORITHMS	ICPC Asia Taichung Regional — Gold Medal · Codeforces Master.
SHIPPING	Live multiplayer browser games (Bullheads, Camelot Nights); AI-agent component of the ExhiBeat interactive exhibition (YTP, 2026).
VESTIGE	Built end to end: perception, World Memory, investigation agent, IncidentLab and the product.

alenhsiao.com · github.com/AlenHsiaoWeiLun

APPENDIX A · HARDWARE INTEGRATION

How the engine maps onto the UGen300.

Stage	On the UGen300 (HailoRT 5)	On the host CPU
Detection	YOLO HEF, NMS on chip, create_infer_model	Letterbox, ByteTrack, zone rules
Speech	genai.Speech2Text (Whisper-Base)	Voice-activity gate, keyword routing
Second opinion	genai.VLM (Qwen2-VL-2B, 336×336 frames)	Masked, overlay-free keyframe; guardrail
Questions	genai.LLM (1.5B)	Retrieval; citations attached by code
Scheduling	One VDevice, group_id "TRACE"	Detection priority; VLM requests queued

Measured finding. Qwen2-VL answered our train-at-platform check correctly only on the clean frame; drawn boxes broke it. The VLM now always sees the masked, unannotated frame.

Memory budget. YOLO, Whisper-Base, Qwen2-VL-2B and a 1.5B LLM must share 8 GB; Stage II week 1 measures residency and switching cost.

Not yet verified on hardware: USB throughput, power under load, multi-model contention. Laptop numbers are labelled as such.

Code: traceai/hailo.py, traceai/perception.py · API per hailo-ai/hailo-apps · details in docs/hailo_port.md

APPENDIX B · TRUST AND PRIVACY

What VESTIGE will not do.

No unauthorized sources

Only cameras and records a workspace is permitted to use. No covert monitoring, no assumed access to campus or public CCTV.

No identities

No face recognition by default. People are pseudonymous tracks; head regions are blurred in stored evidence.

No accusations

Movement is not theft; proximity is not intent. VESTIGE describes observations; people decide responsibility.

No filled gaps

Missing footage stays missing. Synthetic data is labelled and never merged with real observations.

Edge by default

Raw video and audio stay on site; metadata and selected clips are kept for a configurable retention period, then deleted.

Every action audited

Model verdicts and human decisions land in one audit log, aligned with Taiwan's Personal Data Protection Act.

APPENDIX C · SOURCES AND CREDITS

1 · NTU notice on unlocked bicycles being ridden away, 2024-09-13 — politics.ntu.edu.tw/?p=29723

2 · Liberty Times, NTU ≈29,000 bicycles, 2021-02-04 — news.ltn.com.tw/news/life/breakingnews/3432272

3 · West Midlands PCC trial protocol (59 min per 18 min of CCTV), 2020-03-26, citing Davis et al. 2018

4 · MOJ Academy for the Judiciary, 刑事政策與犯罪研究論文集 (23), private CCTV retention

5 · CNA, Taipei Metro briefing on recording hosts, 2025-12-22 — cna.com.tw/news/aloc/202512220295.aspx

6 · 19 Dec 2025 case: Taipei District Prosecutors press release, 2026-01-15; Taipei Metro initial report, 2025-12-19, and statement, 2025-12-22; CNA, 2026-01-15 (3,000+ cameras)

7 · Ministry of Labor, 2025 occupational fatalities, 2026-01-15 — mol.gov.tw/1607/1632/1633/87515/

8 · CNA, reversing road roller fatality, Hualien, 2026-07-31 — cna.com.tw/news/asoc/202607310124.aspx

9 · Verkada, VIVOTEK VORTEX AI-Hub, Ambient.ai Pulsar, Voxel, Protex AI, Intenseye — product pages and press releases, 2025–2026

10 · HK Development Bureau, Smart Site Safety System (4S); TechNews 2026-01-22 (TSMC AI CCTV)

11 · ASUS UGen300 press release 2026-04-01 and UGen Series Sales Kit 2026-09-17

12 · Hailo Model Zoo v5.4 (HAILO10H) and GenAI Model Zoo; hailo-ai/hailo-apps

13 · Hendrycks & Dietterich, ImageNet-C, ICLR 2019; Michaelis et al., Benchmarking Robustness in Object Detection, 2019

14 · Qwen2-VL (2024); Whisper (2022); ByteTrack (ECCV 2022); Ultralytics YOLO11 (2024)

15 · TRACE repository: alert-storm benchmark, IncidentLab reports, stage timings — github.com/AlenHsiaoWeiLun/trace-ugen

16 · Brand note: "VESTIGE" is a provisional name; trademark and domain clearance pending.

IMAGERY AND DATA

Bicycle rack, metro, work site, gate, control room — AI-generated (OpenAI image model via Codex); illustrations, not real places or events.

IncidentLab and alert-storm charts — scripts in the repository, measured 11 Oct 2026; method stated on the page.

Product screenshots (p.05–09) — the running prototype on public stock footage (Pexels, Wikimedia Commons); faces blurred.

UGen300 / Hailo-10H images — ASUS UGen Series Sales Kit (contest material).
Founder photo — personal portrait.

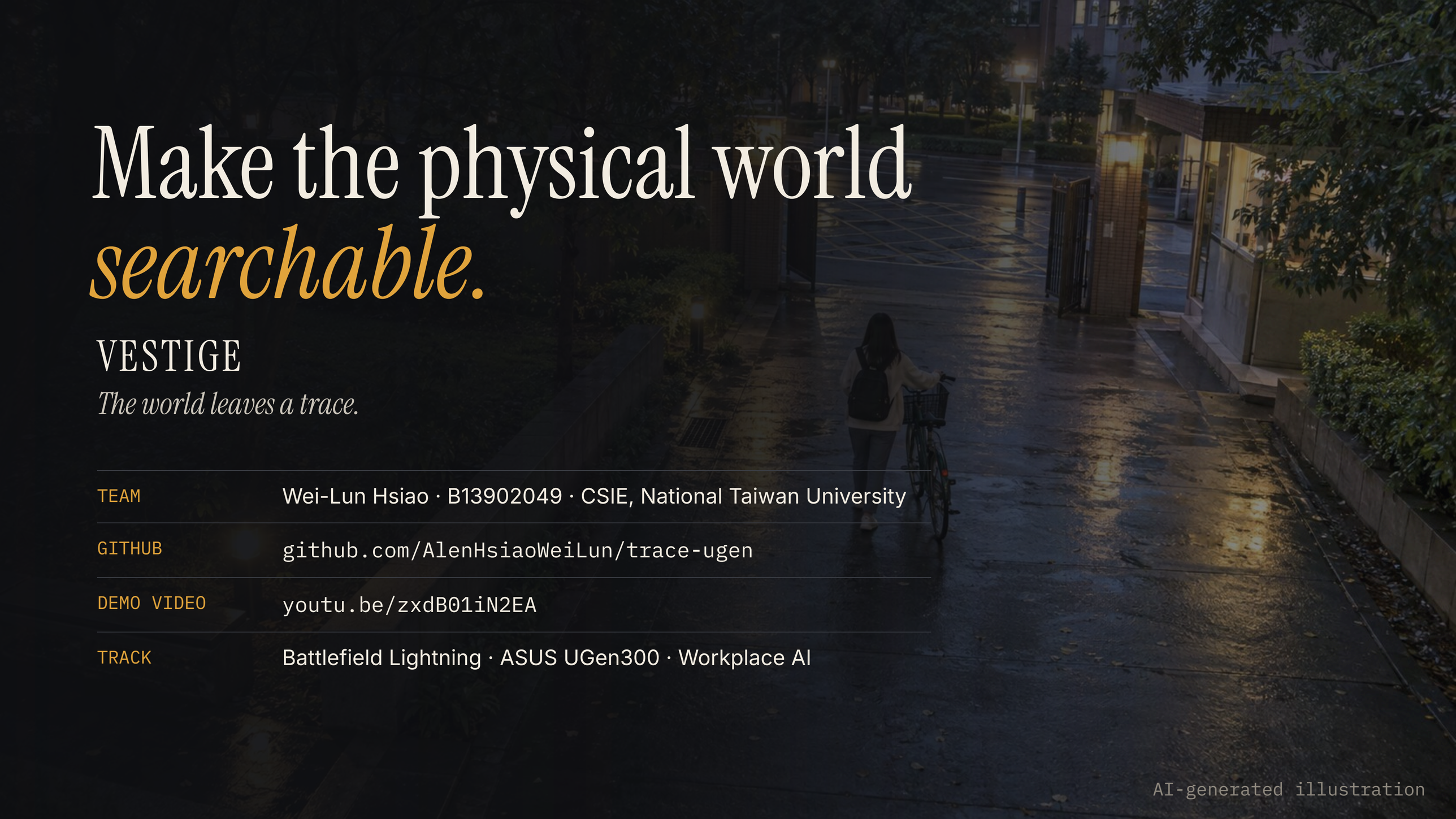
No news photos or real incident footage are used. Development was assisted by AI tools (Claude Code, OpenAI Codex); every number comes from code that can be re-run.

Make the physical world *searchable.*

VESTIGE

The world leaves a trace.

TEAM	Wei-Lun Hsiao · B13902049 · CSIE, National Taiwan University
GITHUB	github.com/AlenHsiaoWeiLun/trace-ugen
DEMO VIDEO	youtu.be/zxdB01iN2EA
TRACK	Battlefield Lightning · ASUS UGen300 · Workplace AI



AI-generated illustration